Chapter 9. Epilogue—Life Outside $\mathbb{R}^n$

Differentiable manifolds and Riemannian manifolds

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Overview

We have dealt with submanifolds of euclidean space and forms on open sets of euclidean space: conceptually simple, but important ideas are sometimes obscured by the familiar surroundings, and in mathematics and physics manifolds often occur as abstract spaces. This chapter sketches how to move outside $\mathbb{R}^n$.

§41. Differentiable Manifolds and Riemannian Manifolds

41.1 Differentiable manifolds

Definition 41.1 (differentiable manifold)

Let M be a metric space. Suppose there is a collection of homeomorphisms $\alpha_i : U_i \to V_i$, where U_i is open in $\mathbb{H}^k$ or $\mathbb{R}^k$ and V_i is open in M, such that the V_i cover M. (α_i is a **homeomorphism** if it carries U_i onto V_i one-to-one and both α_i, α_i^{-1} are continuous.) Suppose the α_i overlap with class C^∞: the **transition function** $\alpha_i^{-1} \circ \alpha_j$ is of class C^∞ whenever $V_i \cap V_j$ is non-empty. The α_i are **coordinate patches** on M, as is any other homeomorphism $\alpha : U \to V$ (U open in $\mathbb{H}^k$ or $\mathbb{R}^k$, V open in M) that overlaps the α_i with class C^∞. M with this collection of patches is a **differentiable k-manifold** (of class C^∞).

For $k = 1$ we again allow domains open in $\mathbb{L}^1$. If there is a patch $\alpha : U \to V$ about p with U open in $\mathbb{R}^k$, p is an **interior point**; otherwise a **boundary point**; the boundary points form ∂M. If $\alpha : U \to V$ is a patch about p, then $p \in \partial M$ iff U is open in $\mathbb{H}^k$ and $p = \alpha(\mathbf{x})$ for some $\mathbf{x} \in \mathbb{R}^{k-1} \times 0$ (proof as in Lemma 24.2). Throughout, M denotes a differentiable k-manifold.

Definition 41.2 (orientation)

Patches α_0, α_1 on M **overlap positively** if $\det D(\alpha_1^{-1} \circ \alpha_0) > 0$. If M can be covered by patches overlapping positively, M is **orientable**. An **orientation** of M consists of such a covering together with all other patches overlapping these positively; an **oriented manifold** is a manifold with an orientation. If $\{\alpha_i\}$ is an orientation, $\{\alpha_i \circ r\}$ (r the reflection of $\mathbb{R}^k$) is the **opposite** orientation.

If $\partial M \neq \emptyset$, then ∂M is a differentiable $(k-1)$-manifold without boundary, with patches $\alpha \circ b$ (α a patch about $p \in \partial M$, $b(x_1, \ldots, x_{k-1}) = (x_1, \ldots, x_{k-1}, 0)$), as in Theorem 24.3. If α_0, α_1 overlap positively, so do $\alpha_0 \circ b$ and $\alpha_1 \circ b$ (Theorem 34.1). So if M is oriented, ∂M can be oriented by restricting the patches of the orientation of M; for k even this is the **induced orientation** of ∂M, for k odd its opposite is.

Definition 41.3 (C^∞ maps between manifolds)

Let M, N be differentiable manifolds of dimensions k, n, $A \subset M$ and $f : A \to N$. f is **of class C^∞** if for each $x \in A$ there are a patch $\alpha : U \to V$ on M about x and a patch $\beta : W \to Y$ on N about $y = f(x)$ such that $\beta^{-1} \circ f \circ \alpha$ is of class C^∞ as a map of a subset of $\mathbb{R}^k$ into $\mathbb{R}^n$. (Independent of the patches, since transition functions are C^∞.) A one-to-one $f : M \to N$ onto N with f and f^{-1} of class C^∞ is a **diffeomorphism**.

41.2 Tangent vectors

Without a surrounding euclidean space, what is a tangent vector? For M in $\mathbb{R}^n$, a tangent vector at p is the velocity $(p; D\gamma(t_0))$ of a C^∞ curve $\gamma : [a,b] \to M$ with $\gamma(t_0) = p$. For abstract M we cannot differentiate γ, but for a patch $\alpha : U \to V$ about p, $\alpha^{-1} \circ \gamma$ maps a subset of $\mathbb{R}^1$ into $\mathbb{R}^k$, so we may think of the "derivative" of γ at t_0 as the function $\mathbf{v}$ assigning to each patch α about p the matrix $\mathbf{v}(\alpha) = D(\alpha^{-1} \circ \gamma)(t_0)$. If α_0, α_1 are two patches about p, the chain rule gives

$$\mathbf{v}(\alpha_1) = Dg(\mathbf{x}_0) \cdot \mathbf{v}(\alpha_0), \qquad g = \alpha_1^{-1} \circ \alpha_0, \quad \mathbf{x}_0 = \alpha_0^{-1}(p).$$

Definition 41.4 (tangent vector, tangent space)

Given $p \in M$, a **tangent vector** to M at p is a function $\mathbf{v}$ assigning to each coordinate patch $\alpha : U \to V$ about p a k by 1 column matrix $\mathbf{v}(\alpha)$, such that for any two patches α_0, α_1 about p,

$$\mathbf{v}(\alpha_1) = Dg(\mathbf{x}_0) \cdot \mathbf{v}(\alpha_0), \tag{$*$}$$

where $g = \alpha_1^{-1} \circ \alpha_0$ and $\mathbf{x}_0 = \alpha_0^{-1}(p)$. The entries of $\mathbf{v}(\alpha)$ are the **components** of $\mathbf{v}$ with respect to α.

By $(*)$, $\mathbf{v}$ is determined by its components in a single patch, and we may define $(a\mathbf{v} + b\mathbf{w})(\alpha) = a\mathbf{v}(\alpha) + b\mathbf{w}(\alpha)$. The tangent vectors at p form the **tangent space** $\mathcal{T}_p(M)$, a k-dimensional vector space: if $\alpha(\mathbf{x}) = p$, the map $\mathbf{v} \mapsto (\mathbf{x}; \mathbf{v}(\alpha))$ is a linear isomorphism of $\mathcal{T}_p(M)$ onto $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k)$, whose inverse is denoted $\alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_p(M)$; so $\alpha_*(\mathbf{x}; \mathbf{v}(\alpha)) = \mathbf{v}$. For a C^∞ curve $\gamma : [a,b] \to M$ with $\gamma(t_0) = p$, the **velocity vector** is defined by $\mathbf{v}(\alpha) = D(\alpha^{-1} \circ \gamma)(t_0)$; every tangent vector at p is the velocity vector of some curve.

Tangent vectors as derivations

A common alternative: to $\mathbf{v} \in \mathcal{T}_p(M)$ associate an operator $X_{\mathbf{v}}$ on real-valued C^∞ functions near p, the **derivative with respect to** $\mathbf{v}$. If $\mathbf{v}$ is the velocity of γ at t_0 and $\alpha(\mathbf{x}) = p$, then writing $f \circ \gamma = (f \circ \alpha) \circ (\alpha^{-1} \circ \gamma)$,

$$\frac{d(f \circ \gamma)}{dt}(t_0) = D(f \circ \alpha)(\mathbf{x}) \cdot D(\alpha^{-1} \circ \gamma)(t_0) = D(f \circ \alpha)(\mathbf{x}) \cdot \mathbf{v}(\alpha),$$

which depends only on f and $\mathbf{v}$. So define $X_{\mathbf{v}}(f) = D(f \circ \alpha)(\mathbf{x}) \cdot \mathbf{v}(\alpha)$; it is independent of α, $X_{\mathbf{v}+\mathbf{w}} = X_{\mathbf{v}} + X_{\mathbf{w}}$, $X_{c\mathbf{v}} = cX_{\mathbf{v}}$, and for $M = \mathbb{R}^k$ it is the directional derivative of f with respect to $\mathbf{v}$. It satisfies (1) *locality*: if f, g agree near p, $X_{\mathbf{v}}(f) = X_{\mathbf{v}}(g)$; (2) *linearity*: $X_{\mathbf{v}}(af + bg) = aX_{\mathbf{v}}(f) + bX_{\mathbf{v}}(g)$; (3) the *product rule*: $X_{\mathbf{v}}(f \cdot g) = X_{\mathbf{v}}(f)g(p) + f(p)X_{\mathbf{v}}(g)$. These characterize $X_{\mathbf{v}}$: every operator X on C^∞ functions near p satisfying (1)–(3) equals $X_{\mathbf{v}}$ for a unique $\mathbf{v} \in \mathcal{T}_p(M)$. So one can define a tangent vector at p to be such an operator; many authors prefer this "intrinsic" definition, which does not involve coordinate patches explicitly.

41.3 Forms on M

Definition 41.5 (forms on a manifold)

An l**-form** on M is a function ω assigning to each $p \in M$ an alternating l-tensor $\omega(p) \in \mathcal{A}^l(\mathcal{T}_p(M))$. If $\alpha : U \to V$ is a patch about p with $\alpha(\mathbf{x}) = p$, let $T = \alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_p(M)$ and $T^* : \mathcal{A}^l(\mathcal{T}_p(M)) \to \mathcal{A}^l(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k))$; define $(\alpha^*\omega)(\mathbf{x}) = T^*(\omega(p))$. ω is **of class** C^∞ **near** p if $\alpha^*\omega$ is of class C^∞ near $\mathbf{x}$ (independent of α), and of class C^∞ if this holds for every patch.

Henceforth all forms are C^∞. $\Omega^l(M)$ is the space of l-forms on M. There are no elementary forms on M giving a canonical form, but $\alpha^*\omega = \sum_{[I]} f_I\, dx_I$ in $\mathbb{R}^k$; the f_I are the **components** of ω with respect to α.

Definition 41.6 (differential)

For an l-form ω on M, $d\omega$ is defined as follows: given $p \in M$ and tangent vectors $\mathbf{v}_1, \dots, \mathbf{v}_{l+1}$ at p, choose a patch α about p with $\alpha(\mathbf{x}) = p$ and set

$$d\omega(p)(\mathbf{v}_1, \dots, \mathbf{v}_{l+1}) = d(\alpha^*\omega)(\mathbf{x})((\mathbf{x}; \mathbf{v}_1(\alpha)), \dots, (\mathbf{x}; \mathbf{v}_{l+1}(\alpha))).$$

That is, pull ω back to $\alpha^*\omega$ in $\mathbb{R}^k$, pull the $\mathbf{v}_i$ back to tangent vectors in $\mathbb{R}^k$, and apply d in $\mathbb{R}^k$. Equivalently (with $\mathbf{a}_i = \mathbf{v}_i(\alpha)$), $d\omega(p)(\alpha_*(\mathbf{x}; \mathbf{a}_1), \dots, \alpha_*(\mathbf{x}; \mathbf{a}_{l+1})) = d(\alpha^*\omega)(\mathbf{x})((\mathbf{x}; \mathbf{a}_1), \dots, (\mathbf{x}; \mathbf{a}_{l+1}))$, i.e. $d\omega$ is the unique $(l+1)$-form on M such that

$$\alpha^*(d\omega) = d(\alpha^*\omega)$$

for every coordinate patch α on M. (The d on the right is the operator in $\mathbb{R}^k$.)

This is independent of the patch, and $d\omega$ is C^∞.

Theorem 41.1

Let M be a compact differentiable manifold. Given a covering of M by coordinate patches, there exist functions $\phi_i : M \to \mathbb{R}$ of class C^∞, for $i = 1, \dots, l$, such that: (1) $\phi_i(p) \geq 0$ for each $p \in M$. (2) For each i, the set $\operatorname{Support} \phi_i$ is covered by one of the given coordinate patches. (3) $\sum \phi_i(p) = 1$ for each $p \in M$.

Proof. Given $p \in M$, choose a patch $\alpha : U \to V$ of the covering about p, $\alpha(\mathbf{x}) = p$, and a non-negative C^∞ function $f : U \to \mathbb{R}$ whose support is compact and contained in U, with $f(\mathbf{x}) > 0$. Define $\psi_p : M \to \mathbb{R}$ by $\psi_p(y) = f(\alpha^{-1}(y))$ for $y \in V$ and $\psi_p(y) = 0$ otherwise. Since $f \circ \alpha^{-1}$ vanishes outside a compact subset of V, ψ_p is C^∞ on M. ψ_p is positive on an open set U_p about p. Cover M by finitely many U_p, for $p = p_1, \dots, p_l$, and set $\lambda = \sum_{j=1}^{l} \psi_{p_j}$ and $\phi_i = (1/\lambda)\psi_{p_i}$. ∎

Definition 41.7 (integral of a form)

Let M be a compact oriented differentiable k-manifold and ω a k-form on M. If $\operatorname{Support} \omega$ lies in a single patch $\alpha : U \to V$ of the orientation, $\int_M \omega = \int_{\operatorname{Int} U} \alpha^*\omega$. In general, with $\phi_1, \dots, \phi_l$ as in Theorem 41.1,

$$\int_M \omega = \sum_{i=1}^{l} \Big[\int_M \phi_i \omega \Big].$$

The usual argument shows this is well-defined and linear.

Theorem 41.2 (Stokes' theorem)

Let M be a compact, oriented differentiable k-manifold. Let ω be a $k-1$ form on M. If ∂M is non-empty, give ∂M the induced orientation; then

$$\int_M d\omega = \int_{\partial M} \omega.$$

If ∂M is empty, then $\int_M d\omega = 0$.

Proof. The proof of Theorem 37.2 goes through verbatim: all computations were carried out within coordinate patches, so no changes are needed. The special conventions for $k = 1$ and ∂M a

0-manifold are handled as before (Theorem 37.4). ■

Chapter 8 also generalizes: the **deRham group** $H^k(M)$ is the quotient of the closed k-forms on M by the exact ones, and there are methods to compute it, including a general *Mayer–Vietoris theorem* relating the deRham groups of $M = U \cup V$, U, V and $U \cap V$ (see Bott–Tu). $H^k(M)$ is obviously a diffeomorphism invariant of M. Strikingly, it is also a *topological* invariant: if M and N are homeomorphic, $H^k(M)$ and $H^k(N)$ are linearly isomorphic. This follows from *deRham's theorem*: the algebra of closed forms modulo exact forms on M is isomorphic to an algebra defined in algebraic topology for any topological space, the "cohomology algebra of M with real coefficients".

41.4 Riemannian manifolds

Not everything generalizes so readily: the volume of M and $\int_M f\,dV$ do not make sense on an abstract differentiable manifold. By §36, the volume of a compact oriented k-manifold M in $\mathbb{R}^n$ is $\mathrm{v}(M) = \int_M \omega_V$, where the volume form ω_V has value 1 on *any orthonormal basis* of $\mathcal{T}_p(M)$ in its natural orientation. This uses the inner product that $\mathcal{T}_p(M) \subset \mathcal{T}_p(\mathbb{R}^n) = p \times \mathbb{R}^n$ inherits from the dot product of $\mathbb{R}^n$; an abstract $\mathcal{T}_p(M)$ has none, so "orthonormal" is meaningless. We need an inner product on each tangent space.

Definition 41.8 (Riemannian metric)

A **Riemannian metric** on a differentiable k-manifold M is an inner product $\langle \mathbf{v}, \mathbf{w} \rangle$ on each tangent space $\mathcal{T}_p(M)$, required to be of class C^∞ as a 2-tensor field on M. A **Riemannian manifold** is a differentiable manifold together with a Riemannian metric.

("Metric" here has nothing to do with "metric space".) Every differentiable manifold has a Riemannian metric (the proof uses a partition of unity), but it is not unique. A Riemannian metric gives a volume function $V(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ on k-tuples in $\mathcal{T}_p(M)$ (as in the exercises of §21), so one defines, for continuous f supported in a single patch $\alpha : U \to V$,

$$\int_M f\,dV = \int_{\mathrm{Int}\,U} (f \circ \alpha)\, V(\alpha_*(\mathbf{x}; \mathbf{e}_1), \ldots, \alpha_*(\mathbf{x}; \mathbf{e}_k)),$$

in general via a partition of unity as in §25, and $\mathrm{v}(M) = \int_M dV$. If M is compact, oriented and Riemannian, $\int_M \omega = \int_M \lambda\,dV$ where $\lambda(p)$ is the value of $\omega(p)$ on an orthonormal k-tuple in $\mathcal{T}_p(M)$ belonging to its natural orientation; if $\lambda \equiv 1$, ω is the **volume form** ω_V of M, and $\mathrm{v}(M) = \int_M \omega_V$.

Many questions arise: the *length* of a smooth curve $\gamma : [a, b] \to M$ is $\int_{t=a}^{t=b} \|\gamma_*(t; \mathbf{e}_1)\|$, the integral of the norm of its velocity vector (using the inner product on $\mathcal{T}_p(M)$); one studies *geodesics* (curves of minimal length joining two points), *curvature*, and so on—the subject of *Riemannian geometry*.

What does not generalize

Most of this book generalizes to abstract differentiable or Riemannian manifolds. The interpretation of Stokes' theorem in terms of scalar and vector fields (§38) does not: the "translation functions" of §31 depend on forms defined in $\mathbb{R}^n$, grad and div apply only to fields in $\mathbb{R}^n$, curl only in $\mathbb{R}^3$, and even the "normal vector" to a manifold depends on the surrounding space. Manifolds, differential forms and Stokes' theorem have meaning outside euclidean space; classical vector analysis does not.